

1 Waiting Period - Person

Utilization

Waiting period processing is an optional function, which allows for complete recording of labor times in shop floor data collection (BDE). This applies to both, gaps between individual personnel postings in BDE as well as between the personnel postings in BDE and the clock-in or clock-out postings of the time & attendance module (PZE).

Even if time & attendance functions are not integrated, a very good posting quality can be achieved in BDE (shop floor data collection) if the advanced logon time function, waiting period processing and appropriate options to log off staff automatically at the end of shifts are in use. This can still be improved for flexible working times if the option "PZE (in/out) controls BDE" is used.

Integration

Provided that the time & attendance component (PZE) is in use, the waiting period processing function can also be controlled by PZE clock-in and clock-out records.

Prerequisite

The waiting period processing is enabled in the basic parameter settings --> BDE --> Waiting period processing BDE --> Activate waiting period processing.

The function can be used in connection with the time & attendance module (PZE). However, using PZE is not mandatory. If integration with PZE is required, the relevant options needs to be enabled in the basic parameter settings --> BDE --> PZE (in/out) controls BDE".

Configuration options

The personal waiting period processing is configured across the system in the basic parameter settings. The document dealing with the basic parameter settings provides further information on this.

Filling the gaps between BDE personnel postings

The "waiting period" is defined in minutes within the basic parameter settings if waiting period processing is enabled. It indicates the maximum time that may pass between a logoff at an operation and the logon to the next operation or it indicates within what time an employee has to log on to an operation after his/her clocking-in. If this waiting period is not exceeded, the person's logon is dated back to the time he/she logged off at last or to his/her clocking-in. In case the waiting period is exceeded, the system posts the time exceeded to the defined waiting period operation.

A "maximum waiting period" is also defined within the "basic parameter settings". The maximum waiting period indicates the period of time as of which HYDRA rejects waiting period postings. If the

maximum waiting period is set to 0, waiting period postings will not be rejected. The system always generates a waiting period posting if waiting period processing is enabled. Even at the end of a working day, it starts with the time when an employee logs off last and ends with the first logon on the next day, provided it is not interrupted by clocking-out if the control by PZE option is enabled. This data record spanning two working days of an employee can be suppressed by the maximum waiting period.

The waiting period has to be posted to a "resource performance account (RPA)". An account has to be defined within the basic parameter settings for posting purposes. Please note that only the account number may be entered, e.g. 9 for RPA U8 or 10 for RPA U9. These accounts are free accounts and can be used for waiting period processing without affecting the events.

Options with "PZE (in/out) controls BDE"



The functions of "PZE controls BDE" process a BREAK clocking (beginning of break) as a clocking-out. The end of the break corresponds to a clocking-in.

No control

Clocking records pertaining to the time & attendance module (PZE) do not result in employees being logged on or off in BDE and do not cause any waiting period processing.

Waiting period processing

Processing is enabled if the waiting period is activated in BDE --> PZE (IN/OUT) controls BDE within the basic parameter settings.

When clocking-in, the person is automatically logged on to a waiting period operation in the shop floor system.

If the user logs on to a production OP during the "waiting period" the duration between the clocking-in and the logon to the production operation is posted onto the registered operation. The logon of the production operation is posted forward to the posting in HYDRA-PZE.

The system generates a "waiting period posting" as soon as the waiting period is exceeded. The data posted to the production order is not changed. This processing is also activated for the times between the logoff and logon of staff during the working time.

Once the person clocks out in the time & attendance module, they are automatically logged off in HYDRA-BDE as well. Provided that the person was no longer logged on to an operation at the time of clocking out, a waiting period posting is generated for the period of time between the last logoff of the person and the clocking-out. This is even the case if the waiting period is not exceeded.

The operation is interrupted automatically if the last person who works on the operation clocks out.

Waiting period processing with "auto in"

Waiting period processing is activated, provided that it is enabled with "auto in" in the basic parameter settings --> BDE --> PZE (IN/OUT) controls BDE. In this case, however, the person is automatically logged on to the operation that was posted the last time by this person.

As a prerequisite for this logon, this operation must neither be active nor finished already.

In case the operation could be logged on, the person is also logged on to the operation. In this case, however, the person is always directly logged on; there is never an advance logon.

The logon is made on that workplace to which the person and the operation were logged on before clocking out. If it is an individual workplace all people who are already active are connected to the new operation and the new person is connected with all OPs that are already active at this workplace.

If an advance logon of staff is required instead, it might be configured by an entry in the INI configuration:

Parameter name	Parameter value
Name	PZE_CONTROLS_BDE
Key	ADVANCED_LOGON_OF_STAFF
Value	TRUE / FALSE
Comment	Generate advance logon of staff optionally

If this processing is activated by the INI entry

- An advance logon of staff is generated instead, when clocking in, provided that the person clocks in within the advance logon period for the machine or the machine is in the "no shift" status (time without shift).
- No OP is logged on, if the person clocks in at a time when the machine is in the "no shift" status and the operation to be posted is in the "automatically interrupted" status.

Waiting period processing with clock-out

If the waiting period processing with clock-out is activated in the basic parameter settings --> BDE --> PZE (IN/OUT) controls BDE, the person is automatically logged off from all machines and orders as soon as they clock out. In addition, the operation is also interrupted, if the last person who is logged on to the operation is logged off.

General notes



Operations are not logged on automatically by the function "PZE controls BDE", provided that resources are posted for the operation or inspection operations are connected with the operation.

In such cases, only the person is logged on or off by the PZE clocking.

When it comes to group workplaces or overhead cost operations, it might also be the case that the person cannot be logged off automatically in the cases mentioned above, as one person always has to be logged on to an operation.

When operations are merged at the terminal, staff is logged off from this merged operation as soon as they clock out. However, they cannot be logged on automatically to the merged operation.

In the mentioned cases, it is recommended to log the operation on and off explicitly, if required.



The "waiting period" and "auto in" options automatically compare the first staff logon within a shift with the rounding rules of HYDRA-PZE if the option "rounding of the first personnel logon" is enabled. This is why, the HYDRA-BDE posting might deviate from the actual time.

If the relevant option is enabled in the basic parameter settings, the personnel logon time in BDE is dated forward to the beginning of the working time temporarily rounded by PZE if a person is logged in by a clocking-in.

Example: a person clocks in at 5.55 a.m. However, the person's working time only starts at 6.00 a.m. according to the shift model and the PZE rounding rules. Waiting period processing logs the person on at 6.00 a.m. in BDE instead at 5.55 a.m. (if enabled at all).

This option only affects such personnel logons in BDE that are performed due to waiting period processing of a person clocking in. This option neither affects staff logging on or off using BDE posting dialogs nor further clocking-in records of the person on the same day.



The waiting period processing is not activated if the person clocks in outside of the skeleton working time, i.e. before the beginning of the skeleton time.



In contrast to HYDRA-BDE, postings in HYDRA-PZE are not transferred ONLINE but at a later point in time. At first they are buffered locally. When it comes to postings that depend on each other (e.g. within the scope of waiting period processing), this might lead to HYDRA-BDE being more up-to-date than HYDRA-PZE postings. In this case, HYDRA-BDE postings are not backdated.



Backdating to the beginning of the last shift at most takes place at workplaces/machines, which are directly assigned to a terminal of the operation mode "MDE".



When the waiting period function for personnel and machines is used at the same time, the person is postdated to the time when the operation is logged on at most. A waiting period posting for the person is generated for the remaining time, irrespective of whether the waiting period has been exceeded or not. The waiting period for machines always takes priority.

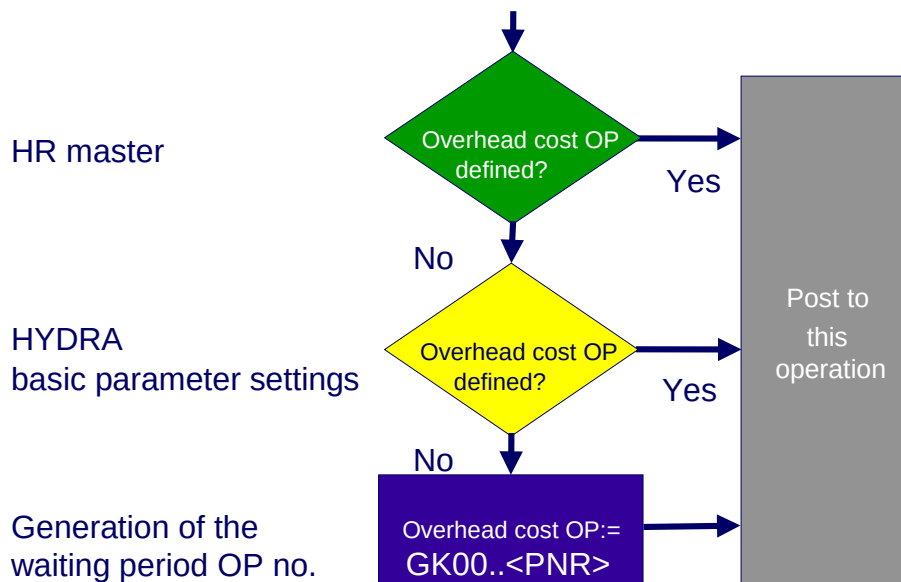


If the person is logged off automatically by clocking out, and if quantities in B records are required, he/she has to upload his/her quantities beforehand using partial uploads.

Effects of the settings in the HR master

The recorded waiting period for a person is compared with the person's annual model as it is defined in the [HR master](#); the generated posting of the record type "B" is assigned the ("master") workplace defined in the HR master. If these fields are not filled out the waiting period is 0, provided that the year model is missing or if the workplace is missing it is set to the value 0 or blank.

Finding of the waiting period OP with waiting period processing based on personnel



The resource performance account that is configured in the basic parameter settings is posted, regardless of the object where the waiting period operation is found.

Upload to the PPS system

By default, waiting period postings are not uploaded to the higher-level ERP system. However, if it is required to upload these postings as well, HYDRA can be customized accordingly. To do so, the HYDRA standard interface (HYD-ERP) has to be used.

Status of a waiting period operation

By default, a waiting period operation is always assigned to the status "available (waiting period)", irrespective of whether the waiting period operation has already been posted or if a person or a machine "is waiting".

Processing is not affected, if the status of a waiting period operation or a waiting period order is changed to the "completed" status using the function "change status" in the order overview or order information dialog. The defined waiting period operation is still posted. Please also take into account the notes on changing a waiting period operation in the basic parameter settings.

Reactivation of a waiting period operation

Processing is not affected if a waiting period operation is reactivated using the "reactivate OP" function in the order overview dialog. By default, the status is set to "available (waiting period)" again in HYDRA (this is an internal status with control indicator "U").

Example 1: Effects of the planned working time in HYDRA-PZE on HYDRA-BDE

Prerequisites:

- HYDRA-BDE and HYDRA-PZE are in use
- The option "PZE (in/out) controls BDE" has to be set to "waiting period " or "Auto in".

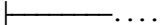
```

Planned attendance PZE Person 1020
                        6:00 am
                        |
Person 1020 clocks in
                        5:55 am
                        |
OP 70010 030 is logged on 6:00 am (determined by the planned attendance)
Person 1020 is logged on |-----
  
```

Example 2 "waiting period exceeded"

The waiting period is set to 10 minutes in this and the following examples

last logoff (or clock-in) of the person
5:30 am



the person is logged on the next time at
6:00 am



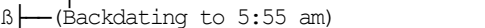
A waiting period posting of 30 minutes is generated for this person and posted on the waiting period RPA for the period of time between 5.30 am and 6.00 am. No backdating takes place.

Example 3 "waiting period not exceeded"

Last logoff or clock-in of the person
5:55 am



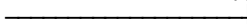
The person is logged in the next time at
6:00 am



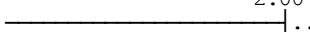
The person's logon is backdated to 5.55 am. The five minutes of waiting period are posted on the waiting period RPA. No waiting period posting is generated.

Example 4 "logoff of staff before clocking out in PZE"

The person is logged off
1:55 pm



The person clocks out
2:00 pm



In this case, a waiting period posting between the last person that logs off (HYDRA-BDE) and the clock-out (HYDRA-PZE) is generated, even if the waiting period is not exceeded, as there is no (longer) an operation onto which this time could be posted. The five minutes of waiting period are posted onto the RPA defined within the basic parameter settings. The time when the last person logs off is not postdated.

Example 5 – "logon of staff with PZE rounding"

The person arrives at 5.55 am. This time is already dated to 6.00 am by the clocking time gradation function of HYDRA-PZE; i.e. the shop floor collection module (BDE) logs the person on at 6.00 am (HYDRA-BDE does not know the original clocking time in HYDRA-PZE). Consequently, the person works on the waiting period OP as of 6.00 am.

Logging an OP/a person on at 6.05 am triggers a backdating or waiting period posting.

As in this case, the waiting period has not been exceeded (6.00 am - 6.05 am: is a maximum of five minutes, which is less than the 10 minutes waiting period), the logon time for the OP/person is backdated to 6.00 am.

If the OP/person is logged on at 6.10.01 am the waiting period is exceeded and HYDRA generates a waiting period posting of 10 minutes and 1 sec.

The logon time for the OP/person remains at 6.10.01 am.